Discrete Mathematics, Algorithms and Applications
(2025) 2550102 (15 pages)
© World Scientific Publishing Company
DOI: 10.1142/S1793830925501022

Ahlswede–Zhang style identities of rough set approximations

Tran Dan Thu

University of Management and Technology, Ho Chi Minh City
No. 2, 60CL Street, Quarter 9, Cat Lai Ward, Thu Duc City
Ho Chi Minh City 700000, Vietnam
thu.tran@umt.edu.vn; tdthu@yahoo.com

Received 9 December 2024
Revised 2 June 2025
Accepted 9 June 2025
Published

Communicated by Bhaskar DasGupta

Rough set theory is a recently developed mathematical framework that has been widely used in big-data analysis and knowledge discovery. In this paper, we establish a novel connection between rough set theory and combinatorial extremal theory by introducing some interesting identities of rough sets of Ahlswede–Zhang style. We prove these identities using combinatorial techniques, providing new insights into the relationship between these two areas. Our results have theoretical implications for rough set theory and its applications in various fields such as data analysis, machine learning, and decision making.

Keywords: Combinatorial identity; AZ-style identity; approximation measure; rough set.

Mathematics Subject Classification 2020: 05D05

1. Introduction

Rough set theory, proposed by Pawlak in the early 1980s [10, 11], is a mathematical framework for knowledge discovery. It divides data of each project into equivalence classes for approximating concepts, called lower approximations and upper approximations. The difference between these two approximations is called the boundary region. In recent years, the theory has become a powerful tool for numerous real-world applications, such as big-data analysis, forecasting, decision support, and disease or software fault prediction [6, 9, 12, 14].

There have been numerous theoretical contributions to deep research on rough sets in different directions. Researchers have proposed mathematical models or extensions of rough sets, such as algebraic methods for the rough set structure [3], topological approaches to the rough approximations [1, 13], and matroid approaches for rough sets [7, 8]. These theoretical contributions have significantly expanded and improved the rough set theory, laying the foundation for its practical applications.

In combinatorial extremal theory, the Ahlswede–Zhang (AZ) identity [2] is well known for sharpening a classical inequality and providing an identity version of the inequality. To establish the identity, Ahlswede and Zhang defined a set function that counts the number of edges of the hypercube $\mathcal{H}_n$ leaving an upset $\mathcal{U} \subseteq \mathcal{H}_n$ from a set that belongs to that upset. In our previous work [15], while exploring possible modifications to the AZ function, we unexpectedly discovered that a slight change to the definition, by formally replacing set intersection with set union, will transform the function into an approximation of rough sets. This unexpected result contributed to giving a connection between combinatorial extremal theory and rough set theory, and led to the establishment of a preliminary identity on rough sets in [15], which we were initially unaware of its significance. This work has represented a valuable contribution to the development of the AZ identity.

Motivated by our previous work [15–17], this paper presents AZ style identities of the lower and upper approximation measures of rough sets, with the goal of establishing a connection between combinatorial extremal theory and rough set theory. In the following section, we introduce and review essential notations and definitions that will be used throughout the paper.

2. AZ Identity and Rough Set Approximations

Let $[n]=\{1, 2, \ldots, n\}$ and let $2^{[n]}$ denote the family of all subsets of $[n]$. For $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, the upset $\mathcal{U}(\mathcal{A})$ generated by $\mathcal{A}$ is defined as

$$\mathcal{U}(\mathcal{A}) = \{U \subseteq [n] : U \supseteq A \text{ for some } A \in \mathcal{A}\}.$$

For each $X \in \mathcal{U}(\mathcal{A})$, Ahlswede and Zhang [2] define the *AZ function* $Z_{\mathcal{A}}(X)$ as

$$Z_{\mathcal{A}}(X) = \bigcap_{X \supseteq A \in \mathcal{A}} A, \tag{2.1}$$

and discovered the following identity, called *AZ identity.*

Theorem 2.1 ([2]). *If $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ and $\emptyset \notin \mathcal{A}$ then*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = 1, \tag{2.2}$$

where $|\cdot|$ denotes the cardinality of a set.

The AZ identity is interesting because it is a sharpening of famous combinatorial inequalities. For example, the Lubell–Yamamoto–Meshalkin (LYM) inequality

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1, \tag{2.3}$$

for any antichain $\mathcal{A} \subseteq 2^{[n]}$, and the Bollobás inequality of two families $\mathcal{A} = \{A_1, A_2, \ldots, A_m\}$ and $\mathcal{B} = \{B_1, B_2, \ldots, B_m\}$ that satisfy $A_i \subseteq B_j$ if and only if $i = j$

$$\sum_{i=1}^{m} \frac{1}{\binom{n-|B_i|+|A_i|}{|A_i|}} \leq 1, \tag{2.4}$$

are sharpened by the AZ identity and its extensions [2, 16, 17].

In rough set theory, two approximation measures $\mathcal{A}^-(X)$ and $\mathcal{A}^+(X)$ of X with respect to a partition $\mathcal{A}$ of $[n]$ are defined in [10] as

$$\mathcal{A}^-(X) = \bigcup_{X \supseteq A \in \mathcal{A}} A, \quad \mathcal{A}^+(X) = \bigcup_{A \in \mathcal{A},\ A \cap X \neq \emptyset} A. \tag{2.5}$$

We have $\mathcal{A}^-(X) \subseteq X \subseteq \mathcal{A}^+(X)$. The sets $\mathcal{A}^-(X)$ and $\mathcal{A}^+(X)$ are, respectively, the lower and upper approximations of the set X. The lower and upper approximations define mathematically precise concepts for data approximations in each project, allowing us to identify and describe key concepts approximately and rigorously. Empty unions, following the convention in set theory, are also empty set. Therefore, $\mathcal{A}^-(X) = \emptyset$ if and only if $X \notin \mathcal{U}(\mathcal{A})$ and $\mathcal{A}^+(X) = \emptyset$ if and only if $X = \emptyset$. It follows that $\emptyset$ is not an element of $\mathcal{A}$ and $\mathcal{A}$ is not empty, because $\mathcal{A}$ is a partition of $[n]$.

Notably, we can obtain the lower approximation $\mathcal{A}^-(X)$ from the AZ function $Z_{\mathcal{A}}(X)$ by formally replacing set intersection with set union. This approach yields the following identity, which is quite similar to the AZ identity but for rough set approximation [15]

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} = |\mathcal{A}|, \tag{2.6}$$

for an arbitrary partition $\mathcal{A}$ of $[n]$. This identity shows a constraint between rough set approximations, which will be investigated more in the subsequent sections.

2.1. *Rough set approximation measures*

The positive, negative, and boundary regions of the set X are defined, respectively, as follows [11, 20]:

$$\mathrm{POS}_{\mathcal{A}}(X) = \mathcal{A}^-(X), \quad \mathrm{NEG}_{\mathcal{A}}(X) = \overline{\mathcal{A}^+(X)}, \quad \mathrm{BND}_{\mathcal{A}}(X) = \mathcal{A}^+(X) \backslash \mathcal{A}^-(X),$$

where $\overline{B}$ denotes the complement set of B in $[n]$, i.e., $\overline{B} = [n] \backslash B$ for $B \subseteq [n]$. The positive region of X, $\mathrm{POS}_{\mathcal{A}}(X)$, consists of elements that are definitely members of X. And the negative region, $\mathrm{NEG}_{\mathcal{A}}(X)$, consists of elements that are definitely not members of X. On the other hand, the boundary region, $\mathrm{BND}_{\mathcal{A}}(X)$, consists of elements whose membership in X is uncertain. Obviously, we have

$$|\mathrm{POS}_{\mathcal{A}}(X)| = |\mathcal{A}^-(X)|, \quad |\mathrm{NEG}_{\mathcal{A}}(X)| = n - |\mathcal{A}^+(X)|,$$
$$|\mathrm{BND}_{\mathcal{A}}(X)| = |\mathcal{A}^+(X)| - |\mathcal{A}^-(X)|.$$

To quantify the accuracy with which a set X is approximated by its positive and negative regions, Yao defines two complementary approximation measures for X as follows [20]:

$$\gamma_{\mathcal{A}}(X) = \frac{|\mathrm{POS}_{\mathcal{A}}(X)| + |\mathrm{NEG}_{\mathcal{A}}(X)|}{n}; \quad \beta_{\mathcal{A}}(X) = \frac{|\mathrm{BND}_{\mathcal{A}}(X)|}{n}. \tag{2.7}$$

It readily follows that these measures satisfy $\gamma_{\mathcal{A}}(X) = 1 - \frac{|\mathrm{BND}_{\mathcal{A}}(X)|}{n}$ and, consequently, $\beta_{\mathcal{A}}(X) = 1 - \gamma_{\mathcal{A}}(X)$.

Furthermore, the accuracy of approximation of X in $\mathcal{A}$ is defined by Pawlak [10, 11] as

$$\alpha_{\mathcal{A}}(X) = \frac{|\mathcal{A}^{-}(X)|}{|\mathcal{A}^{+}(X)|} = \frac{|\mathrm{POS}_{\mathcal{A}}(X)|}{n - |\mathrm{NEG}_{\mathcal{A}}(X)|}, \tag{2.8}$$

and the relative number of elements in X which can be approximated by the partition $\mathcal{A}$ is defined by Gediga and Düntsch [5] as

$$\pi_{\mathcal{A}}(X) = \frac{|\mathcal{A}^{-}(X)|}{|X|}, \tag{2.9}$$

where $X \neq \emptyset$ to avoid undefined results.

Clearly, $0 \leq \gamma_{\mathcal{A}}(X), \beta_{\mathcal{A}}(X) \leq 1$ and $0 \leq \alpha_{\mathcal{A}}(X) \leq \pi_{\mathcal{A}}(X) \leq 1$ for $X \neq \emptyset$. The following proposition is a direct consequence of the definition of the two measures $\pi_{\mathcal{A}}(\cdot)$ and $\alpha_{\mathcal{A}}(\cdot)$. It characterizes the part of the space $2^{[n]}$ where these two measures are equal to zero.

Proposition 2.2. *For all $\emptyset \neq X \subseteq [n]$, we have*

$$\pi_{\mathcal{A}}(X) = 0 \Leftrightarrow \alpha_{\mathcal{A}}(X) = 0 \Leftrightarrow X \notin \mathcal{U}(\mathcal{A}).$$

2.2. *Relationships between well-known classical metrics and rough set approximation measures*

Yao [19] and Gediga and Düntsch [5] introduced an inverse measure of the measure $\pi_{\mathcal{A}}(X)$, which can be interpreted using the well-known Marczewski–Steinhaus metric. The distance between two sets A and B is defined as $d(A, B) = \frac{\mu(A \Delta B)}{\mu(A \cup B)}$. When the measure $\mu(X)$ is $\mu(X) = |X|$, the Marczewski–Steinhaus metric becomes

$$d_J(A, B) = \frac{|A \Delta B|}{|A \cup B|} = \frac{|A \cup B| - |A \cap B|}{|A \cup B|} = 1 - \frac{|A \cap B|}{|A \cup B|}, \tag{2.10}$$

which is the Jaccard metric itself.

Then, the inverse measure is defined as $\sigma_{\mathcal{A}}(X) = d_J(X, \mathcal{A}^{-}(X))$. Since $\mathcal{A}^{-}(X) \subseteq X$, we have

$$\sigma_{\mathcal{A}}(X) = d_J(X, \mathcal{A}^{-}(X)) = 1 - \frac{|\mathcal{A}^{-}(X)|}{|X|} = 1 - \pi_{\mathcal{A}}(X) \in [0, 1].$$

We can see that the Jaccard distance $d_J(A,B)$ can be expressed in terms of the Hamming distance $d_H(A,B) = |A\Delta B|$ as

$$d_J(A,B) = \frac{d_H(A,B)}{|A\cup B|}. \tag{2.11}$$

Similarly to $\sigma_{\mathcal{A}}(\cdot)$, the rough set measures $\gamma_{\mathcal{A}}(\cdot)$ and $\alpha_{\mathcal{A}}(\cdot)$ can be formulated using the Jaccard distance as follows:

$$\gamma_{\mathcal{A}}(X) = d_J(\mathrm{BND}_{\mathcal{A}}(X), [n]), \tag{2.12}$$

and

$$\alpha_{\mathcal{A}}(X) = 1 - d_J(\mathcal{A}^-(X), \mathcal{A}^+(X)). \tag{2.13}$$

Many properties of these measures have been extensively studied by Gediga and Düntsch [5], Yao [19], and Wang *et al.* [18]. Despite extensive research on rough set theory, no identities have yet been discovered for approximation measures. In the following sections, we present novel identities of AZ style for approximation measures and prove them using combinatorial techniques.

3. AZ Style Identities of Rough Set Approximations

3.1. *Two identities of the lower and the upper approximation*

Let H_r denote the rth harmonic number, where $H_0 = 0$ and $H_r = \sum_{k=1}^{r}\frac{1}{k}$ for $r \geq 1$.

Proposition 3.1. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} = |\mathcal{A}|, \tag{3.1}$$

and

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^+(X)|}{|X|\binom{n}{|X|}} = \sum_{A\in\mathcal{A}} |A| H_{|A|}. \tag{3.2}$$

The identity (3.1) was originally introduced in [15] (see Theorem 5 and its proof therein). Engel provides an alternative proof of this identity (see [4, Sec. 8]), highlighting its deep connection to the AZ identity. A proof of these two identities will be given in Sec. 4. In the following sections, we present identities involving approximation measures of rough sets. These identities are immediate consequences of Proposition 3.1. However, direct proofs without using this crucial proposition are not easy.

3.2. *Identities of the measure $\pi_{\mathcal{A}}(X)$ and their properties*

Regarding the measure $\pi_{\mathcal{A}}(X)$, the following theorem presents an initial combinatorial identity for approximation measures of rough sets.

Theorem 3.2. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{\pi_{\mathcal{A}}(X)}{\binom{n}{|X|}} = |\mathcal{A}|. \tag{3.3}$$

Proof. Due to the definition of the measure $\pi_{\mathcal{A}}(X)$, $\pi_{\mathcal{A}}(X) = \frac{|\mathcal{A}^{-}(X)|}{|X|}$, the identity (3.3) is equivalent to $\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^{-}(X)|}{|X|\binom{n}{|X|}} = |\mathcal{A}|$, namely the identity (3.1) in Proposition 3.1. $\square$

Because $\emptyset \notin \mathcal{U}(\mathcal{A})$ and $\pi_{\mathcal{A}}(X) = 0$ if $X \notin \mathcal{U}(\mathcal{A})$, the summation in the left-hand side of the identity (3.3) is only over sets X that are in $\mathcal{U}(\mathcal{A})$. Therefore, the identity is equivalent to

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{\pi_{\mathcal{A}}(X)}{\binom{n}{|X|}} = |\mathcal{A}|. \tag{3.4}$$

Let k be an integer, $1 \leq k \leq n$, the average value of $\pi_{\mathcal{A}}(X)$ over all subsets X of size k of $[n]$ is defined as $P_k(\mathcal{A}) = \frac{1}{\binom{n}{k}} \sum_{X \subseteq [n],\ |X|=k} \pi_{\mathcal{A}}(X)$. Due to the identity (3.3), the average value of $P_k(\mathcal{A})$ for $k = 1, 2, \ldots, n$ is $\frac{1}{n}\sum_{k=1}^{n} P_k(\mathcal{A}) = \frac{|\mathcal{A}|}{n}$. Consequently, this value is estimated by $\frac{|\mathcal{A}|}{n} \in \mathbf{O}(1)$ since $0 < \frac{|\mathcal{A}|}{n} \leq 1$. Moreover, this value tends to be close to zero as practical applications since the cardinality of $\mathcal{A}$ is much smaller than n.

The following result presents an identity of the measure $\sigma_{\mathcal{A}}(\cdot)$, which is the same style as that of the measure $\pi_{\mathcal{A}}(X)$.

Theorem 3.3. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{\sigma_{\mathcal{A}}(X)}{\binom{n}{|X|}} = n - |\mathcal{A}|. \tag{3.5}$$

Proof. For all nonempty set $X \subseteq [n]$, $\sigma_{\mathcal{A}}(X) + \pi_{\mathcal{A}}(X) = 1$. This implies that

$$\begin{aligned}\sum_{\emptyset \neq X \subseteq [n]} \frac{\sigma_{\mathcal{A}}(X) + \pi_{\mathcal{A}}(X)}{\binom{n}{|X|}} &= \sum_{\emptyset \neq X \subseteq [n]} \frac{1}{\binom{n}{|X|}} = \sum_{k=1}^{n} \sum_{X \subseteq [n], |X|=k} \frac{1}{\binom{n}{|X|}} \\ &= \sum_{k=1}^{n} \binom{n}{k} \frac{1}{\binom{n}{k}} = n,\end{aligned}$$

and consequently, the identity (3.5) follows directly from the identity (3.3). $\square$

Especially, when $\mathcal{A}$ consists of singletons, both sides of identities (3.3) and (3.5) are n and zero, respectively. This is an extremal case where the measures $\pi_{\mathcal{A}}(\cdot)$ and $\sigma_{\mathcal{A}}(\cdot)$ are maximized and minimized, respectively, as shown in the following proposition. Its proof derives easily from the above definitions.

Proposition 3.4. *The following statements are equivalent*:

(i) *$\mathcal{A}$ consists of singletons, i.e.*, $|\mathcal{A}| = n$;
(ii) *For all* $\emptyset \neq X \subseteq [n]$, $\mathcal{A}^-(X) = X$;
(iii) *For all* $\emptyset \neq X \subseteq [n]$, $X = \mathcal{A}^+(X)$;
(iv) *For all* $\emptyset \neq X \subseteq [n]$, $\sigma_{\mathcal{A}}(X) = 0$;
(v) *For all* $\emptyset \neq X \subseteq [n]$, $\pi_{\mathcal{A}}(X) = 1$;
(vi) *For all* $\emptyset \neq X \subseteq [n]$, $\alpha_{\mathcal{A}}(X) = 1$.

The average value similar to the average value of $P_k(\mathcal{A})$, with $\pi_{\mathcal{A}}(\cdot)$ replaced by $\sigma_{\mathcal{A}}(\cdot)$, is $\frac{n-|\mathcal{A}|}{n} = 1 - \frac{|\mathcal{A}|}{n}$, which approximates to 1 when the cardinality of $\mathcal{A}$ is much smaller than n.

3.3. *Identities of the measures $\beta_{\mathcal{A}}(X)$ and $\gamma_{\mathcal{A}}(X)$*

Lemma 3.5. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{\gamma_{\mathcal{A}}(X)}{|X|\binom{n}{|X|}} + \sum_{\emptyset\neq X\subseteq[n]} \frac{\beta_{\mathcal{A}}(X)}{|X|\binom{n}{|X|}} = H_n. \tag{3.6}$$

Proof. For all $\emptyset \neq X \subseteq [n]$, $\beta_{\mathcal{A}}(X) + \gamma_{\mathcal{A}}(X) = 1$, this enables us to simplify the expression on the left-hand side of (3.6) to

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{1}{|X|\binom{n}{|X|}} = \sum_{k=1}^{n} \binom{n}{k} \frac{1}{k\binom{n}{k}} = \sum_{k=1}^{n} \frac{1}{k} = H_n. \qquad \square$$

The identities of the two approximation measures $\beta_{\mathcal{A}}(\cdot)$ and $\gamma_{\mathcal{A}}(\cdot)$ involve terms containing harmonic numbers as shown in the following theorem.

Theorem 3.6. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{\beta_{\mathcal{A}}(X)}{|X|\binom{n}{|X|}} = \frac{1}{n}\sum_{A\in\mathcal{A}} |A|H_{|A|} - \frac{|\mathcal{A}|}{n}, \tag{3.7}$$

and

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{\gamma_{\mathcal{A}}(X)}{|X|\binom{n}{|X|}} = H_n + \frac{|\mathcal{A}|}{n} - \frac{1}{n}\sum_{A\in\mathcal{A}} |A|H_{|A|}. \tag{3.8}$$

Proof. By Lemma 3.5, identities (3.7) and (3.8) are immediately equivalent. To prove (3.7), we rewrite its left-hand side by substituting the definition $\beta_{\mathcal{A}}(X) = \frac{|\mathcal{A}^+(X)|-|\mathcal{A}^-(X)|}{n}$, to yield

$$\text{LHS} = \sum_{\emptyset\neq X\subseteq[n]} \frac{\beta_{\mathcal{A}}(X)}{|X|\binom{n}{|X|}} = \frac{1}{n}\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^+(X)|}{|X|\binom{n}{|X|}} - \frac{1}{n}\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}}.$$

Applying Proposition 3.1 to each sum in the last expression directly produces the right-hand sides of (3.7). $\square$

3.4. *Identities involving the positive, negative, and boundary regions*

In this section, we introduce AZ style identities for the approximations of rough sets $\mathrm{POS}_{\mathcal{A}}(X)$, $\mathrm{BND}_{\mathcal{A}}(X)$, and $\mathrm{NEG}_{\mathcal{A}}(X)$. These identities are established by using Proposition 3.1.

Lemma 3.7. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} + \sum_{Y \subsetneq [n]} \frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}} = nH_n, \tag{3.9}$$

and

$$\sum_{Y \subsetneq [n]} \frac{|\mathcal{A}^-(Y)|}{(n-|Y|)\binom{n}{|Y|}} + \sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^+(X)|}{|X|\binom{n}{|X|}} = nH_n. \tag{3.10}$$

Proof. Since $|\mathcal{A}^+(Y)| = n - |\mathcal{A}^-(\overline{Y})|$ (see Properties of Approximations [10]) and $\binom{n}{|Y|} = \binom{n}{n-|Y|}$, we can rewrite the second sum in the left-hand side of (3.9) to obtain

$$\sum_{Y \subsetneq [n]} \frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}} = \sum_{Y \subsetneq [n]} \frac{n - |\mathcal{A}^-(\overline{Y})|}{(n-|Y|)\binom{n}{n-|Y|}} = \sum_{\emptyset \neq X \subseteq [n]} \frac{n - |\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}},$$

which yields

$$\begin{aligned} &\sum_{Y \subsetneq [n]} \frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}} + \sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} \\ &\quad = \sum_{\emptyset \neq X \subseteq [n]} \frac{n}{|X|\binom{n}{|X|}} = n \sum_{k=1}^{n} \binom{n}{k} \frac{1}{k\binom{n}{k}} = nH_n. \end{aligned}$$

Hence, the identity (3.9) holds. The identity (3.10) can be proved similarly. $\square$

AZ style identities typically involve terms of the forms $|X|\binom{n}{|X|}$ or $(n-|Y|)\binom{n}{|Y|}$. The following proposition presents the identities of this style for the positive regions of rough sets.

Proposition 3.8. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathrm{POS}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = |\mathcal{A}|, \tag{3.11}$$

and

$$\sum_{Y \subsetneq [n]} \frac{|POS_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}} = nH_n - \sum_{A \in \mathcal{A}} |A| H_{|A|}. \tag{3.12}$$

Proof. The identity (3.11) follows directly from Proposition 3.1 due to the definitions of the positive regions, $|\mathrm{POS}_{\mathcal{A}}(X)| = |\mathcal{A}^-(X)|$. To prove (3.12), we apply Lemma 3.7 to evaluate its left-hand side as

$$\sum_{Y \subsetneq [n]} \frac{|\mathrm{POS}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}} = \sum_{Y \subsetneq [n]} \frac{|\mathcal{A}^-(Y)|}{(n-|Y|)\binom{n}{|Y|}} = nH_n - \sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^+(X)|}{|X|\binom{n}{|X|}}.$$

Then, using Proposition 3.1 again to finish the proof. □

The identities for boundary regions differ from those for positive regions, where both right-hand sides of the identities are the same, as given in the following proposition.

Proposition 3.9. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathrm{BND}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \sum_{A \in \mathcal{A}} |A| H_{|A|} - |\mathcal{A}|, \tag{3.13}$$

and

$$\sum_{Y \subsetneq [n]} \frac{|\mathrm{BND}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}} = \sum_{A \in \mathcal{A}} |A| H_{|A|} - |\mathcal{A}|. \tag{3.14}$$

Proof. The identity (3.13) can be directly derived from identity (3.7) in Theorem 3.6 by utilizing the property $|\mathrm{BND}_{\mathcal{A}}(X)| = n\beta_{\mathcal{A}}(X)$. Moreover, by applying Lemma 3.7, we obtain $\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathrm{BND}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \sum_{Y \subsetneq [n]} \frac{|\mathrm{BND}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}}$, and then conclude that the identity (3.14) also holds. □

The negative regions have an equality relation with the positive regions through terms of the forms $|X|\binom{n}{|X|}$ or $(n-|Y|)\binom{n}{|Y|}$.

Proposition 3.10. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{Y \subsetneq [n]} \frac{|\mathrm{NEG}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}} = \sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathrm{POS}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}}, \tag{3.15}$$

and

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathrm{NEG}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \sum_{Y \subsetneq [n]} \frac{|\mathrm{POS}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}}. \tag{3.16}$$

Proof. We rewrite the left-hand side of (3.15) as

$$\begin{aligned}\sum_{Y\subsetneq[n]}\frac{|\mathrm{NEG}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}} &= \sum_{Y\subsetneq[n]}\frac{n-|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}}\\ &= n\sum_{Y\subsetneq[n]}\frac{1}{(n-|Y|)\binom{n}{|Y|}}-\sum_{Y\subsetneq[n]}\frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}}\\ &= n\sum_{k=1}^{n}\binom{n}{k}\frac{1}{k\binom{n}{k}}-\sum_{Y\subsetneq[n]}\frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}}\\ &= nH_n-\sum_{Y\subsetneq[n]}\frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}}.\end{aligned}$$

Then, applying the identity (3.9) in Lemma 3.7 to obtain its right-hand side

$$nH_n-\sum_{Y\subsetneq[n]}\frac{|\mathcal{A}^+(Y)|}{(n-|Y|)\binom{n}{|Y|}}=\sum_{\emptyset\neq X\subseteq[n]}\frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}}=\sum_{\emptyset\neq X\subseteq[n]}\frac{|\mathrm{POS}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}}.$$

The identity (3.16) can be proved similarly. □

Corollary 3.11. *If $\mathcal{A}$ is a partition of $[n]$ then*

$$\sum_{Y\subsetneq[n]}\frac{|\mathrm{NEG}_{\mathcal{A}}(Y)|}{(n-|Y|)\binom{n}{|Y|}}=|\mathcal{A}|, \tag{3.17}$$

and

$$\sum_{\emptyset\neq X\subseteq[n]}\frac{|\mathrm{NEG}_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}}=nH_n-\sum_{A\in\mathcal{A}}|A|H_{|A|}. \tag{3.18}$$

4. Proofs of Proposition 3.1

This section provides direct proofs of the two identities in Proposition 3.1. An important elementary identity in the following lemma is used. This elementary identity will be used to simplify combinatorial expressions when proving lemmas and propositions.

Lemma 4.1 ([16, Lemma 1]). *Let a,b,c be integers such that $a\geq 0$, $b>0$, $c\geq a+b$. Then*

$$\sum_{j=0}^{a}\binom{a}{j}\frac{1}{(b+j)\binom{c}{b+j}}=\frac{1}{b\binom{c-a}{b}}. \tag{4.1}$$

While seemingly elementary, the proof of Lemma 4.1 requires technical details that preclude a brief exposition. Its complete proof can be found in [16].

Lemma 4.2. *If* $\emptyset \neq A \subseteq [n]$ *then*

$$\sum_{A \subseteq X \subseteq [n]} \frac{1}{|X|\binom{n}{|X|}} = \frac{1}{|A|}. \tag{4.2}$$

Proof. By observing all possibilities of $X \subseteq [n]$ such that $X \supseteq A$, the left-hand side of (4.2) is

$$\text{LHS} = \sum_{j=0}^{n-|A|} \binom{n-|A|}{j} \frac{1}{(|A|+j)\binom{n}{|A|+j}} = \sum_{j=0}^{a} \binom{a}{j} \frac{1}{(b+j)\binom{c}{b+j}},$$

where $a = n - |A| \geq 0$, $b = |A| > 0$, $c = n = a + b$. Then, applying Lemma 4.1, the left-hand side becomes

$$\text{LHS} = \frac{1}{b\binom{c-a}{b}} = \frac{1}{|A|\binom{n-n+|A|}{|A|}} = \frac{1}{|A|},$$

which is the right-hand side of(4.2). □

The case of summation over X where $X \subseteq [n]$ and $X \cap A \neq \emptyset$ requires a lemma analogous to Lemma 4.2.

Lemma 4.3. *If* $\emptyset \neq A \subseteq [n]$ *then*

$$\sum_{X \subseteq [n], X \cap A \neq \emptyset} \frac{1}{|X|\binom{n}{|X|}} = H_{|A|}. \tag{4.3}$$

Proof. The left-hand side of (4.3) is rewritten as

$$\sum_{k=1}^{|A|} \binom{|A|}{k} \sum_{j=0}^{n-|A|} \binom{n-|A|}{j} \frac{1}{(k+j)\binom{n}{k+j}} = \sum_{k=1}^{|A|} \binom{|A|}{k} T_n(|A|),$$

where $T_n(a) = \sum_{j=0}^{n-a} \binom{n-a}{j} \frac{1}{(k+j)\binom{n}{k+j}}$.

Since $n - |A|$, k, n satisfy the condition in Lemma 4.1 on a, b, c, respectively, we apply Lemma 4.1 to obtain $T_n(a) = \sum_{j=0}^{n-a} \binom{n-a}{j} \frac{1}{(k+j)\binom{n}{k+j}} = \frac{1}{k\binom{n-(n-a)}{k}} = \frac{1}{k\binom{a}{k}}$. Therefore, the left-hand side is equal to $\sum_{k=1}^{|A|} \binom{|A|}{k} T_n(|A|) = \sum_{k=1}^{|A|} \binom{|A|}{k} \frac{1}{k\binom{|A|}{k}} = H_{|A|}$. □

4.1. *Proofs of the identity* (3.1) *of Proposition* 3.1

If $\mathcal{A}$ *is a partition of* $[n]$ *then*

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} = |\mathcal{A}|. \tag{3.1}$$

Proof. Since $\mathcal{A}$ is a partition of $[n]$, any two sets in $\mathcal{A}$ are disjoint. It follows that

$$|\mathcal{A}^-(X)| = \left|\bigcup_{X\supseteq A\in\mathcal{A}} A\right| = \sum_{X\supseteq A\in\mathcal{A}} |A|.$$

Using this equality, we can rewrite the left-hand side of (3.1)

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^-(X)|}{|X|\binom{n}{|X|}} = \sum_{\emptyset\neq X\subseteq[n]} \frac{\sum_{X\supseteq A\in\mathcal{A}}|A|}{|X|\binom{n}{|X|}} = \sum_{\emptyset\neq X\subseteq[n]}\sum_{X\supseteq A\in\mathcal{A}} \frac{|A|}{|X|\binom{n}{|X|}}$$

$$= \sum_{A\in\mathcal{A}}\sum_{A\subseteq X\subseteq[n]} \frac{|A|}{|X|\binom{n}{|X|}} = \sum_{A\in\mathcal{A}} |A| \sum_{A\subseteq X\subseteq[n]} \frac{1}{|X|\binom{n}{|X|}}.$$

At this point, we can apply the identity (4.2) to the last sum

$$\sum_{A\in\mathcal{A}} |A|\frac{1}{|A|} = \sum_{A\in\mathcal{A}} 1 = |\mathcal{A}|.$$

This completes the proof of the identity (3.1). □

4.2. *Proofs of the identity* (3.2) *of Proposition* 3.1

If $\mathcal{A}$ is a partition of $[n]$ then

$$\sum_{\emptyset\neq X\subseteq[n]} \frac{|\mathcal{A}^+(X)|}{|X|\binom{n}{|X|}} = \sum_{A\in\mathcal{A}} |A|H_{|A|}. \qquad (3.2)$$

Proof. Following the same approach as the previous proof, substituting the value

$$|\mathcal{A}^+(X)| = \left|\bigcup_{X\subseteq[n],X\cap A\neq\emptyset} A\right| = \sum_{X\subseteq[n],X\cap A\neq\emptyset} |A|,$$

into the left-hand side of the identity (3.2), then applying Lemma 4.3 to obtain the right-hand side

$$\sum_{A\in\mathcal{A}}\sum_{X\subseteq[n],X\cap A\neq\emptyset} \frac{|A|}{|X|\binom{n}{|X|}} = \sum_{A\in\mathcal{A}} |A| \sum_{X\subseteq[n],X\cap A\neq\emptyset} \frac{1}{|X|\binom{n}{|X|}} = \sum_{A\in\mathcal{A}} |A|H_{|A|}. \qquad \square$$

5. Practical Applications of Rough Set Identities

In complex data environments, a key challenge lies in comprehensively measuring how effectively attributes definitively identify any possible concept or group of objects. Let $U = [n]$ be the universe of objects. To address this, we may use a holistic index for the certain definability of knowledge, encompassing every potential concept within the data. This ideal index can be represented by the following sum:

$$\sum_{\emptyset \neq X \subseteq U} \frac{\pi_{\mathcal{A}}(X)}{\binom{n}{|X|}},$$

in which $\pi_{\mathcal{A}}(X) = \frac{|\mathcal{A}^{-}(X)|}{|X|}$ is the internal accuracy of each $X \subseteq U$, $X \neq \emptyset$. The sum theoretically quantifies the aggregate discriminative power of attributes to precisely identify and define groups of objects within the system. The normalization factor $\binom{n}{|X|}$ balances contributions across subsets of the same size, ensuring a comprehensive evaluation across all potential concepts.

The direct computation of this theoretically profound index is, however, practically infeasible. With n objects in U, the sum involves 2^n terms. For instance, if n is just 256, the number of terms (2^{256}) is a computationally intractable number. This renders direct calculation impossible for real-world datasets, where n can be significantly larger. Such exponential growth has historically prevented the practical use of the index, severely limiting insights into overall knowledge quality.

The identity (3.3) introduces a solution for a significant computational barrier. This otherwise intractable sum precisely equals the cardinality of equivalence classes generated by the indiscernibility relation on U, which is also known to be $|\mathcal{A}|$. This identity provides a powerful tool for accurately assessing overall knowledge quality without exponential computation. Its significant practical applications include facilitating effective computation in large-scale information systems.

Similarly, these identities concerning rough set approximation measures reveal a fundamental link between the detailed, micro-level aspects of information and its broader, macro-level structure. Their left-hand sides represent extensive sums that become computationally unmanageable as the number of objects grows, making direct calculation impractical for real-world data. Conversely, their right-hand sides offer quantities that are straightforward to compute, directly reflecting essential properties of the underlying information system, such as the count and size distribution of knowledge granules. This connection provides a practical way to assess complex data quality and discriminative power, offering valuable tools for knowledge system design and analysis.

6. Conclusion

Unlike the AZ identity [2], which applies to an antichain, rough set identities are designed for partitions. A commonality between the two approaches arises when each partition collapses to a single element, $\mathcal{A} = \{A\}$, which also forms a trivial antichain. Extension approaches in rough set theory do not impose the requirement that $\mathcal{A}$ must be a partition. An interesting direction for future research is to explore identities within extension models of rough sets where $\mathcal{A}$ is not a partition.

This paper introduces several rough set identities that bridge the gap between combinatorial theory and rough set theory. These identities reveal quantitative characteristics of rough set approximation measures. A promising direction is to examine variants of these identities in extension or generalization models of rough sets, such as covering rough set models [14], topological generalizations [1, 13], or matroid research for rough sets [7, 8]. Additionally, investigating the implications of these identities in their respective fields is crucial. Last but not least, assessing the practical utility of these identities in real-world problems is essential [3, 6, 14].

ORCID

Tran Dan Thu https://orcid.org/0000-0001-9094-9356

References

[1] M. Abo-Elhamayel and Y. Yang, Generalizations of rough sets via topology, *Afrika Mat.* **32** (2021) 41–50.

[2] R. Ahlswede and Z. Zhang, An identity in combinatorial extremal theory, *Adv. Math.* **80** (1990) 137–151.

[3] S. Akama, Y. Kudo and T. Murai, Overview of rough set theory, in *Topics in Rough Set Theory*, Intelligent Systems Reference Library (ISRL), Vol. 168 (Springer, 2020), pp. 7–60.

[4] K. Engel and D. T. Tran, Boundary optimization for rough sets, *Discrete Math.* **341**(9) (2018) 2465–2477.

[5] G. Gediga and I. Düntsch, Rough approximation quality revisited, *Artif. Intell.* **132** (2001) 219–234.

[6] F. Jiang, Q. Hu, Z. Yang, J. Liu and J. Du, A neighborhood rough sets-based ensemble method, with application to software fault prediction, *Expert Syst. Appl.* **264** (2025) 125919.

[7] Y. Liu, H. Zhao and W. Zhu, Parametric matroid of rough set, *Internat. J. Uncertain. Fuzziness Knowledge-Based Systems* **23**(6) (2015) 893–908.

[8] V. W. Marek and A. Skowron, Rough sets and matroids, in *Transactions on Rough Sets XVII*, Lecture Notes in Computer Science, Vol. 8375 (Springer, 2014).

[9] S. Naouali, S. B. Salem and Z. Chtourou, Clustering categorical data: A survey, *Int. J. Inf. Technol. Decis. Mak.* **19**(1) (2020) 49–96.

[10] Z. Pawlak, Rough sets, *Int. J. Comput. Inf. Sci.* **11** (1982) 341–356.

[11] Z. Pawlak, *Rough Sets*: *Theoretical Aspects of Reasoning About Data* (Kluwer Academic Publishers, 1991).

[12] P. Pieta and T. Szmuc, Applications of rough sets in big data analysis: An overview, *Int. J. Appl. Math. Comput. Sci.* **31**(4) (2021) 659–683.

[13] P. K. Singh and S. Tiwari, Topological structures in rough set theory: A survey, *Hacet. J. Math. Stat.* **49**(4) (2020) 1270–1294.

[14] A. Skowron and S. Dutta, Rough sets: Past, present, and future, *Nat. Comput.* **17** (2018) 855–876.

[15] D. T. Tran, Identities for combinatorial extremal theory, *Bull. London Math. Soc.* **29** (1997) 693–696.

[16] D. T. Tran, An AZ-style identity and Bollobás deficiency, *J. Combin. Theory Ser. A* **114**(8) (2007) 1504–1514.

[17] D. T. Tran, AZ-style identities of downsets and upsets, *Discrete Math. Algorithms Appl.* **6**(3) (2014) 1450040.

[18] Z. Wang, X. Zhang and J. Deng, The uncertainty measures for covering rough set models, *Soft Comput.* **24** (2020) 11909–11929.

[19] Y. Y. Yao, Information granulation and rough set approximation, *Int. J. Intell. Syst.* **16** (2001) 87–104.

[20] Y. Y. Yao, Notes on rough set approximations and associated measures, *J. Zhejiang Ocean Univ. Nat. Sci.* **29**(5) (2010) 399–410.